

Traffic Light Controller Project Report

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Introduction

The objective of this project was to design, program, and physically implement a sophisticated Traffic Light Control System using the TM4C123GH6PM ARM Cortex-M4 microcontroller. Unlike a simple timing loop, this system was engineered to simulate a real-world intersection that prioritizes both vehicle flow and pedestrian safety through low-level hardware-software integration.

By leveraging the power of ARM Assembly, the project focused on creating a high-performance environment where time-critical tasks—such as a 1-second countdown and an emergency pedestrian interrupt—occur seamlessly without halting the primary system logic.

The primary mission of this project was to bridge the gap between abstract digital logic and physically embedded systems. The key goals include:

- **Finite State Machine (FSM) Optimization:** Developing a robust, label-based FSM architecture to manage transitions between Green, Yellow, and Red states. The goal was to ensure the system remains in a predictable state at all times, preventing logic "fall-through" and ensuring reliable traffic cycles.
- **Interrupt-Driven Concurrency:** Implementing two distinct types of hardware interrupts:
 - **Periodic Interrupts (SysTick):** Utilizing the internal 24-bit SysTick timer to provide a precise 1-second heartbeat for the system, replacing inefficient software delay loops with hardware-timed accuracy.

- Edge-Triggered Interrupts (GPIO): Configuring an external tactile button on a breadboard to trigger a Rising Edge interrupt on Port F, allowing for an immediate, asynchronous response to a pedestrian crossing request.
- Hardware Interfacing and Circuit Design: Constructing an external circuit on a breadboard to interface with the microcontroller. This involved:
 - Wiring a custom-mapped 7-segment display (SSD) to Port B to provide a visual 10-second countdown for pedestrians.
 - Configuring a tri-color LED array on Port F to represent traffic signals.
 - Applying electrical engineering principles such as current-limiting resistors and pull-down configurations to ensure signal integrity.
- Low-Level Resource Management: Maximizing the efficiency of the TM4C123's limited resources by writing high-performance ARM Assembly. This included direct manipulation of Memory-Mapped I/O registers and managing SRAM variables for shared data synchronization between the main FSM and the interrupt handlers.

At its core, this project serves as a demonstration of a modern embedded controller.

Under normal operation, the system provides a steady 30-second throughput for vehicles. Upon a pedestrian interrupt, the system intelligently transitions to a specialized "Pedestrian Yellow" state, characterized by a flashing hazard light and a synchronized numerical countdown on the SSD, ensuring that human safety is integrated directly into the machine's primary logic